Matrix multiplication

$$A \cdot B = ?$$

- more complicated
- motivation will become clear later

Product only defined if # columns in A = # rows in B

$$\underset{\substack{A\\ m\times \underline{n}}}{\begin{pmatrix} \quad & \quad \end{pmatrix}}\underset{\substack{B\\ \underline{p}\times q}}{\begin{pmatrix} \quad & \quad \end{pmatrix}} = \underset{\substack{C\\ m\times q}}{\begin{pmatrix} \quad & \quad \end{pmatrix}}$$

need n = p

Example

$$\begin{pmatrix} a_{11} & a_{12} & a_{13} & a_{14} \\ a_{21} & a_{22} & a_{23} & a_{24} \\ a_{31} & a_{32} & a_{33} & a_{34} \end{pmatrix} \begin{pmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \\ b_{31} & b_{32} \\ b_{41} & b_{42} \end{pmatrix} = \begin{pmatrix} c_{11} & c_{12} \\ c_{21} & c_{22} \\ c_{31} & c_{32} \end{pmatrix}$$

A B C

$3 \times \underline{4}$ equal $\underline{4} \times 2$ 3×2

dot product

$$\begin{aligned} c_{11} &= (\text{row } 1 \text{ of } A) \cdot (\text{column } 1 \text{ of } B) \\ &= a_{11}b_{11} + a_{12}b_{21} + a_{13}b_{31} + a_{14}b_{41} \\ &= \sum_{k=1}^{4} a_{1k}b_{k1} \end{aligned}$$

$$\begin{aligned} c_{12} &= (\text{row } 1 \text{ of } A) \cdot (\text{column } 2 \text{ of } B) \\ &= a_{11}b_{12} + a_{12}b_{22} + a_{13}b_{32} + a_{14}b_{42} \\ &= \sum_{k=1}^{4} a_{1k}b_{k2} \end{aligned}$$

$$c_{21} = (\text{row } 2 \text{ of } A) \cdot (\text{column } 1 \text{ of } B)$$

Example

$$\begin{pmatrix} 3 & 4 \\ 2 & 2 \end{pmatrix} \begin{pmatrix} 1 & 2 \\ -1 & 1 \end{pmatrix} = \begin{pmatrix} -1 & 10 \\ 0 & 6 \end{pmatrix}$$

Check your calculations with
Octave/Matlab or similar!

Matrix multiplication is not commutative!

I.e., in general $AB \neq BA$

To show that matrix multiplication is not commutative, one counterexample is enough.

Take for instance:

$$\underset{A}{\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}} \underset{B}{\begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}} = \begin{pmatrix} 0 & 0 \\ 1 & 1 \end{pmatrix}$$

not equal

But

$$\underset{B}{\begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}} \underset{A}{\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}} = \begin{pmatrix} 1 & 1 \\ 0 & 0 \end{pmatrix}$$

Other example:

$$\underset{\substack{A\\ 1\times 3}}{\begin{pmatrix}1 & 2 & 3\end{pmatrix}}\underset{\substack{B\\ 3\times 1}}{\begin{pmatrix}1\\2\\3\end{pmatrix}} = \underset{1\times 1}{\begin{pmatrix}14\end{pmatrix}}$$

not equal

But

$$\underset{\substack{B\\ 3\times 1}}{\begin{pmatrix}1\\2\\3\end{pmatrix}}\underset{\substack{A\\ 1\times 3}}{\begin{pmatrix}1 & 2 & 3\end{pmatrix}} = \underset{3\times 3}{\begin{pmatrix}1 & 2 & 3\\2 & 4 & 6\\3 & 6 & 9\end{pmatrix}}$$

Always keep the order when multiplying matrices!

Theorem For all matrices A, B, and C,

(1) $A(B+C) = AB + AC$ left distributive law

check on your own!

(2) $(A+B)C = AC + BC$ right distributive law

check on your own!

(3) $A(BC) = (AB)C$ associative law

Proof (1) and (2) are relatively straightforward..

Proof of (3) (associativity):

Let A be $m \times n$, B be $n \times p$, and C be $p \times q$.
We want to show that $A(BC) = (AB)C$.

To this end, we show that for all i, j :

$$(A(BC))_{ij} = ((AB)C)_{ij}$$

entry in row i, column j of (AB)C

By definition of matrix multiplication, we have

$$(A(BC))_{ij} = \sum_{k=1}^{n} a_{ik} \underbrace{\left(\sum_{\ell=1}^{p} b_{k\ell} c_{\ell j} \right)}_{(BC)_{kj}} = \sum_{k=1}^{n} \sum_{\ell=1}^{p} a_{ik} b_{k\ell} c_{\ell j}$$

$$= \sum_{\ell=1}^{p} \sum_{k=1}^{n} a_{ik} b_{k\ell} \underline{c_{\ell j}} = \sum_{\ell=1}^{p} c_{\ell j} \underbrace{\left(\sum_{k=1}^{n} a_{ik} b_{k\ell} \right)}_{(AB)_{i\ell}}$$

$$= \sum_{\ell=1}^{p} \underbrace{\left(\sum_{k=1}^{n} a_{ik} b_{k\ell} \right)}_{(AB)_{i\ell}} c_{\ell j}$$

$$= ((AB)C)_{ij}$$

$\square$

Now let's see various important "alternative ways" of carrying out matrix multiplication.

First, let's look at the extremely important special case where one of the two matrices is a vector.

a.) "matrix · column vector"

Example $\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix} \begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix}$ = column vector

A $\quad$ x

by definition,

$$A\mathbf{x} = \begin{pmatrix} 1\cdot 1 + (-1)\cdot 2 + 2\cdot 3 \\ 1\cdot 4 + (-1)\cdot 5 + 2\cdot 6 \\ 1\cdot 7 + (-1)\cdot 8 + 2\cdot 9 \end{pmatrix} = \begin{pmatrix} 5 \\ 11 \\ 17 \end{pmatrix}$$

equal

Often, the following way of doing the multiplication is more convenient:

$$A\mathbf{x} = 1\begin{pmatrix} 1 \\ 4 \\ 7 \end{pmatrix} + (-1)\begin{pmatrix} 2 \\ 5 \\ 8 \end{pmatrix} + 2\begin{pmatrix} 3 \\ 6 \\ 9 \end{pmatrix} = \begin{pmatrix} 5 \\ 11 \\ 17 \end{pmatrix}$$

Here, Ax is written as a linear combination (weighted sum) of the columns of A. The entries in x determine how the columns are weighted in the weighted sum.

This view of "matrix * column vector" multiplication is particularly convenient if there are (many) zero entries in x:

Example

$$\underset{A}{\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}} \underset{x}{\begin{pmatrix} 1 \\ -1 \\ 0 \end{pmatrix}} = 1 \begin{pmatrix} 1 \\ 4 \\ 7 \end{pmatrix} + (-1) \begin{pmatrix} 2 \\ 5 \\ 8 \end{pmatrix} = \begin{pmatrix} -1 \\ -1 \\ -1 \end{pmatrix}$$

"First column - second column"

Here even more useful:

Example

$$\underset{A}{\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}} \underset{x}{\begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}} = \begin{pmatrix} 3 \\ 6 \\ 9 \end{pmatrix}$$

multiplication just "grabs the third column of A"

b.) What about "matrix · row vector"?

Example

$$\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix} \begin{pmatrix} 2 & 7 & -9 \end{pmatrix} =$$

undefined!

c.) "row vector · matrix"

Example

$$\underbrace{\begin{pmatrix} 1 & -1 & 0 \end{pmatrix}}_{y} \underbrace{\begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix}}_{A} = \begin{pmatrix} -3 & -3 & -3 \end{pmatrix}$$

equal

We can view this multiplication as a linear combination (weigthed sum) of the rows of A:

$$1 \cdot \begin{pmatrix} 1 & 2 & 3 \end{pmatrix} + (-1) \cdot \begin{pmatrix} 4 & 5 & 6 \end{pmatrix} + 0 \cdot \begin{pmatrix} 7 & 8 & 9 \end{pmatrix} = \begin{pmatrix} -3 & -3 & -3 \end{pmatrix}$$

d.) What about "column vector · matrix"?

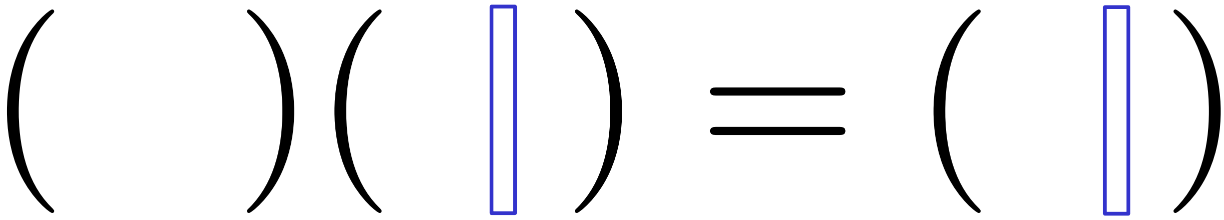
Example

$$\begin{pmatrix} 2 \\ -3 \\ 5 \end{pmatrix} \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix} =$$

undefined!

General matrix multiplication (two arbitrary matrices)

Column by column



A $m \times n$ B $n \times p$ C $m \times p$

We can determine the product "column by column" with p "matrix * column vector" multiplications:

$$\boxed{\text{"column } j \text{ of } C\text{"} = A \cdot \text{"column } j \text{ of } B\text{"}}$$

Example

$$\underset{A}{\begin{pmatrix} 1 & -1 & 2 \\ 3 & 4 & 5 \end{pmatrix}} \underset{B}{\begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 1 \end{pmatrix}} = \underset{C}{\begin{pmatrix} & ? & \end{pmatrix}}$$

First column of the product:

$$\begin{pmatrix} 1 & -1 & 2 \\ 3 & 4 & 5 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} = 1 \begin{pmatrix} 1 \\ 3 \end{pmatrix} + 0 \begin{pmatrix} -1 \\ 4 \end{pmatrix} + 0 \begin{pmatrix} 2 \\ 5 \end{pmatrix} = \begin{pmatrix} 1 \\ 3 \end{pmatrix}$$

first column of A

Second column of the product:

$$\begin{pmatrix} 1 & -1 & 2 \\ 3 & 4 & 5 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \\ 1 \end{pmatrix} = 0 \begin{pmatrix} 1 \\ 3 \end{pmatrix} + 1 \begin{pmatrix} -1 \\ 4 \end{pmatrix} + 1 \begin{pmatrix} 2 \\ 5 \end{pmatrix} = \begin{pmatrix} 1 \\ 9 \end{pmatrix}$$

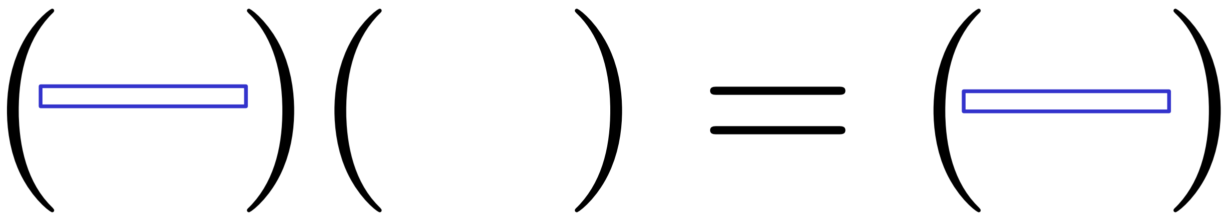
sum of last two columns of A

Hence $\begin{pmatrix} 1 & -1 & 2 \\ 3 & 4 & 5 \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 3 & 9 \end{pmatrix}$

We see that the columns of C are linear combinations (weighted sums) of the columns of A

We can also compute row-by row:

Row by row



A $m \times n$ B $n \times p$ C $m \times p$

We can determine the product "row by row" with m "row vector * matrix" multiplications:

$$\boxed{\text{"row } i \text{ of } C\text{"} = \text{"row } i \text{ of } A\text{"} \cdot B}$$

Example
$$\underset{A}{\begin{pmatrix}1 & -1 & 2\\ 3 & 4 & 5\end{pmatrix}}\underset{B}{\begin{pmatrix}1 & 0\\ 0 & 1\\ 0 & 1\end{pmatrix}} = \underset{C}{\begin{pmatrix} & ? & \end{pmatrix}}$$

First row of the product:

$$\begin{pmatrix}1 & -1 & 2\end{pmatrix}\begin{pmatrix}1 & 0\\ 0 & 1\\ 0 & 1\end{pmatrix} = 1\begin{pmatrix}1 & 0\end{pmatrix} + (-1)\begin{pmatrix}0 & 1\end{pmatrix} + 2\begin{pmatrix}0 & 1\end{pmatrix} = \begin{pmatrix}1 & 1\end{pmatrix}$$

Second row of the product:

$$\begin{pmatrix}3 & 4 & 5\end{pmatrix}\begin{pmatrix}1 & 0\\ 0 & 1\\ 0 & 1\end{pmatrix} = 3\begin{pmatrix}1 & 0\end{pmatrix} + 4\begin{pmatrix}0 & 1\end{pmatrix} + 5\begin{pmatrix}0 & 1\end{pmatrix} = \begin{pmatrix}3 & 9\end{pmatrix}$$

Hence $\begin{pmatrix}1 & -1 & 2\\ 3 & 4 & 5\end{pmatrix}\begin{pmatrix}1 & 0\\ 0 & 1\\ 0 & 1\end{pmatrix} = \begin{pmatrix}1 & 1\\ 3 & 9\end{pmatrix}$

We see that the rows of C are linear combinations (weighted sums) of the rows of B

Identity matrices

Definition

The identity matrix I_n is an $n \times n$ matrix with 1 on the diagonal and 0 everywhere else.

$$I_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \qquad I_3 = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \qquad \text{etc.}$$

Fact If A is an $m \times n$ matrix then

$$I_m A = A = A I_n.$$

Example

$$\underset{A}{\begin{pmatrix} 2 & -1 & 1 \\ 5 & 4 & 7 \end{pmatrix}} \underset{I_3}{\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}} = \underset{A}{\begin{pmatrix} 2 & -1 & 1 \\ 5 & 4 & 7 \end{pmatrix}}$$

Identity matrices of different sizes needed.

$$\underset{I_2}{\begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}} \underset{A}{\begin{pmatrix} 2 & -1 & 1 \\ 5 & 4 & 7 \end{pmatrix}} = \underset{A}{\begin{pmatrix} 2 & -1 & 1 \\ 5 & 4 & 7 \end{pmatrix}}$$

Check using any of the matrix multiplication "methods"!
Which one is most convenient in each case?

Sometimes Useful:

Fact If AB is defined then $(AB)^T = B^T A^T$.

Change of order!

Proof: Tutorial 1